CHAPTER 53

Common Psychiatric Diseases

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Depression

Psychiatric disorders

1. These are characterised by the emotion of anxiety, worrisome thoughts, avoidance behaviour and the somatic symptoms of autonomic arousal.
2. The nature and prominence of the somatic symptoms often lead the patient to present initially to medical services.
3. Anxiety may be stress related and phobic anxiety may follow an unpleasant incident.
4. Patients with anxiety often also have depression.
5. Anxiety disorders are divided into three main subtypes: phobic, paroxysmal (panic) and generalised

Classification of anxiety disorders

	Phobic anxiety disorder	Panic disorder	Generalised anxiety disorder
Occurrence	Situational	Paroxysmal	Persistent
Behaviour	Avoidance	Escape	Agitation
Cognitions	Fear of situation	Fear of symptoms	Worry
Symptoms	On exposure	Episodic	Persistent

Phobic anxiety disorder

1. A phobia is an abnormal or excessive fear of an object or situation, which leads to avoidance of it (such as excessive fear of dying in an air crash leading to avoidance of flying).
2. A generalised phobia of going out alone or being in crowded places is called agoraphobia.
3. Phobic responses can develop to medical interventions such as venepuncture.

Panic disorder

1. Panic disorder describes repeated attacks of severe anxiety, which are not restricted to any particular situation or circumstances.
2. Somatic symptoms such as chest pain, palpitations and paraesthesiae in lips and fingers are common.
3. The symptoms are in part due to involuntary over-breathing (hyperventilation).
4. Patients often fear they are suffering from a serious illness such as a heart attack or stroke, and may therefore seek emergency medical attention.
5. Panic disorder is often associated with agoraphobia.

Generalised anxiety disorder

1. This is chronic anxiety associated with uncontrollable worry.
2. Somatic symptoms of muscle tension and bowel disturbance often lead to a medical presentation.

Obsessive-compulsive disorder

1. Obsessive-compulsive disorder (OCD) is characterised by obsessive thoughts, which are recurrent, unwanted and usually anxiety provoking, and by compulsions, which are repeated acts performed to relieve feelings of tension.
2. An example is repeated hand-washing because of thoughts of contamination.
3. The differential diagnosis is from normal checking behaviour and from delusional beliefs about thought possession.
4. OCD is equally common in men and women.

Mood or effective disorders

Mood or affective disorders include:

- *Unipolar depression:* one or more episodes of low mood and associated symptoms
- *Bipolar disorder:* episodes of elevated mood interspersed with episodes of depression

- *dysthymia:* chronic low-grade depressed mood without sufficient other symptoms to count as clinically significant or major depression.

Depression

1. Major depressive disorder has a prevalence of 5% in the general population and approximately 10% in chronically ill medical out patients.
2. It is a major cause of disability and of suicide.
3. If comorbid with a medical condition, depression magnifies disability, diminishes adherence to medical treatment and rehabilitation, and may even shorten life expectancy.
4. Such comorbid depression may incrementally worsen health more than any combination of chronic diseases without depression.

Aetiology

1. There is a genetic predisposition to depression, especially when of early onset.
2. The number and identity of the genes are largely unknown but the serotonin transporter gene is a candidate.
3. Adversity and emotional deprivation early in life also predispose to depression.
4. Depressive episodes are often triggered by stressful life events (especially those that involve loss), including medical illnesses.
5. Associated biological factors include hypofunction of monoamine neurotransmitter systems (5-HT and noradrenaline (norepinephrine)) and abnormal hypothalamo-pituitary-adrenal axis (HPA) regulation, which results in elevated cortisol levels that do not suppress with dexamethasone.

Diagnosis

1. Depression may be mild, moderate or severe, and episodic, recurrent or chronic.
2. Pointers to an organic cause for psychiatric disorder
   - Late age of onset of psychiatric illness
   - No previous history of psychiatric illness
   - No family history of psychiatric illness
   - No apparent psychological precipitant

Depressed Mood

Presenting Problems in Psychiatric Illness

Anxiety symptoms–

1. Anxiety may be transient, persistent, episodic or limited to specific situations.
2. Most commonly it is transient, as an adjustment disorder and subsides with time.
3. Anxiety may occasionally be a manifestation of a medical condition such as thyrotoxicosis

Symptoms of anxiety disorder

Psychological

Apprehension	Fear of impending disaster
Irritability	Poor concentration
Worry	Depersonalisation

Somatic

Palpitations	Frequent desire to pass urine
Fatigue	Chest pain
Tremor	Initial insomnia
Dizziness	Breathlessness
Sweating	Headache
Diarrhoea	

Differential diagnosis of anxiety

- Normal response to threat
- Adjustment disorder
- Generalised anxiety disorder
- Panic disorder
- Phobic disorder
- Organic (medical) cause
- Hyperthyroidism
- Paroxysmal arrhythmias
- Pheochromocytoma
- Alcohol and benzodiazepine withdrawal
- Hypocalcaemia
- Temporal lobe epilepsy

Depressed mood

1. Depressive disorder is common, occurring in a quarter to a half of medical in patients.
2. The symptoms of depression are both mental and physical.

3. In the physically ill patient, diagnosis of comorbid depression is based mainly on careful assessment of the core psychological symptoms: namely, persistently lowered mood and anhedonia.

Symptoms of depressive disorders

Psychological

- Depressed mood
- Loss of interest
- Reduced self-esteem
- Loss of enjoyment (anhedonia)
- Pessimism
- Guilt
- Suicidal thinking

Somatic

- Reduced appetite
- Loss of libido
- Weight change
- Bowel disturbance
- Disturbed sleep
- Motor retardation (slowing of activity)
- Fatigue

Differential diagnosis

1. Depressive disorder must be differentiated from an adjustment disorder with depressed mood
2. Adjustment disorders are common, self-limiting reactions to adversity, including physical illness, which are transient and require only general support.
3. On the other hand, depressive disorders are characterised by more severe and persistent mood disturbance and do require specific treatment.
4. In some cases, depression may occur as a result of a direct effect of a medical condition or its treatment on neurotransmitters or neural pathways in the brain, when it is referred to as an 'organic mood disorder.

Organic affective disorders

Neurological Infections :

- Cerebrovascular disease
- Infectious mononucleosis
- Cerebral tumour
- Herpes simplex
- Multiple sclerosis
- Brucellosis
- Parkinson's disease
- Typhoid
- Huntington's disease
- Toxoplasmosis
- Alzheimer's disease

Connective tissue disease

- Epilepsy
- Systemic lupus erytematosus

Endocrine	Drugs
• Hypothyroidism	• Reserpine, methyldopa
• Hyperthyroidism	• Phenothiazines
• Cushing's syndrome	• Phenylbutazone
• Addison's disease	• Corticosteroids, oral contraceptives
• Hyperparathyroidism	
Malignant disease	• Interferon

Suicide

1. Depression is the major risk factor for suicide, especially when it co-occurs with a medical condition or substance misuse.
2. When depression is suspected, tactful enquiry should always be made into suicidal thoughts and plans.
3. Asking about suicide does not increase the risk of it occurring, whereas failure to enquire denies the opportunity to prevent it.

Risk factors for suicide

- Psychiatric illness (depressive illness, schizophrenia)
- Older age
- Male sex
- Living alone
- Unemployed
- Recently bereaved, divorced or separated
- Chronic physical ill health
- Drug or alcohol misuse
- Suicide note written

- History of previous attempts (especially if a violent method was used)

Elated mood

1. Elation, or euphoria, is the converse of depression and is characteristic of mania.
2. It may manifest as infectious joviality, over-activity, lack of sleep and appetite, undue optimism, over-talkativeness, irritability, and recklessness in spending and sexual behaviour.
3. When more severe, psychotic symptoms are often evident, such as delusional degrees of grandiosity.
4. Elevated mood is much less common than depression, and where it arises in medical settings is often secondary to drug or alcohol misuse, an organic disorder or medical treatment.

Delusions and hallucinations

Delusions

1. Various types of delusion are identified on the basis of their content.
2. They may be:
   - Persecutory, such as a conviction that others are out to get one
   - Hypochondriacal, such as an unfounded conviction that one has cancer
   - Grandiose, such as a belief that one has special powers or status.
3. The most extreme form of delusion is nihilistic, e.g. 'My head is missing; 'I have no body; 'I am dead.' Delusions should be differentiated from overvalued ideas, which are strongly held but not fixed.

Hallucinations

1. These are perceptions without external stimuli. They can occur in any sensory modality, most commonly visual or auditory.
2. Typical examples are hearing voices when no one else is present, or seeing 'visions'.
3. Hallucinations have the quality of ordinary perceptions and are perceived as originating in the external world, not the patient's own mind (when they are termed pseudo-hallucinations).
4. Those occurring when falling asleep ('hypnagogic') and on waking ('hypnopompic') are not pathological.
5. Hallucinations should be distinguished from illusions, which are misperceptions of real external stimuli (such as mistaking a shrub for a person in poor light).

Anxiety neurosis

Introduction and Management

Anxiety is a normal reaction to stress and can actually be beneficial in some situations. For some people, however, anxiety can become excessive. Anxiety disorders are a group of mental disorders characterized by feelings of anxiety and fear. Anxiety is a worry about forthcoming events and fear is a response to present events. The term anxiety covers four aspects of experiences that the person may have; mental uneasiness, physical pressure, physical symptoms and dissociative anxiety. These feelings may cause physical symptoms, such as a palpitation and trembling.

The emotions present in anxiety disorders range from simple nervousness to bouts of terror. Other psychiatric and medical problems may mimic with the symptoms of an anxiety disorder such as hyperthyroidism. Anxiety disorders occur about twice as often in females as males, and generally begin during childhood.

Classification

There are several forms of anxiety disorders. According to Diagnostic and Statistical Manual of Mental Disorders, 4th Edition, Text Revision, also known as DSM-IV-TR, a manual published by the American Psychiatric Association (APA) that includes all currently recognized mental health disorders, the following is the classification of Anxiety disorders.

1. Generalized anxiety disorder
2. Panic disorder with agoraphobia
3. Panic disorder without agoraphobia
4. Agoraphobia without history of panic disorder
5. Specific phobia

6. Social phobia
7. Obsessive-compulsive disorder
8. Posttraumatic stress disorder
9. Acute stress disorder
10. Anxiety disorder due to a general medical condition
11. Anxiety disorder NOS

Apart from the above, there are some other conditions, which are clinically found. They are Separation Anxiety, Situational Anxiety.

Generalized Anxiety Disorder (Gad)

Generalized anxiety disorder is the most common anxiety disorder to affect adults. It is a common chronic disorder characterized by long-lasting anxiety that is not concentrated on any one object or situation. Those suffering from generalized anxiety disorder experience non-specific persistent fear and worry, and become overly concerned with everyday matters.

Generalized anxiety disorder is "characterized by chronic excessive worry accompanied by three or more of the following symptoms: restlessness, fatigue, concentration problems, irritability, muscle tension, and sleep disturbance". Anxiety can be a symptom of a medical or substance abuse problem, and medical professionals must be aware of this.

Clinical presentation : The patient looks anxious, with increased sweating from the hands, feet, and axillae,and he/she may be weeping, which can suggest depression. Before a diagnosis of anxiety disorder is made, the physician must rule out drug-induced anxiety and other medical causes. In children GAD may be associated with headaches, restlessness, abdominal pain, and heart palpitations. Typically, it begins around 8 to 9 years of age.

Diagnosis: A diagnosis of GAD is made when an individual has been disproportionately anxious about an everyday problem for six months or more. A person may find that he/she has problems making daily decisions and remembering commitments as a result of lack of concentration/ preoccupation with worry.

Panic Disorder: In panic disorder, an individual suffers from brief attacks of extreme terror and anxiety, often marked by trembling, shaking, confusion, dizziness, nausea, and/or difficulty breathing. These panic attacks, defined by the APA as fear or discomfort that abruptly arises and peaks in less than ten minutes, can last for several hours. Attacks can be triggered by stress, fear, or even exercise; the specific cause is not always apparent.

Clinical presentation

As such, those suffering from panic disorder experience symptoms even outside specific panic episodes. Often, normal changes in heartbeat are noticed by a panic sufferer, leading them to think something is wrong with their heart or they are about to have another panic attack. In some cases, a heightened awareness (hypervigilance) of body functioning occurs during panic attacks, wherein any perceived physiological change is interpreted as a possible life-threatening illness.

Diagnosis : A diagnosis of panic disorder requires that said attacks have chronic consequences: either worry over the attacks' potential implications, persistent fear of future attacks, or significant changes in behavior related to the attacks.

Agoraphobia

Agoraphobia is the specific anxiety about being in a place or situation where escape is difficult or uncomfortable or where help may be unavailable. Agoraphobia is strongly linked with panic disorder and is often precipitated by the fear of having a panic attack.

Clinical presentation : A common expression involves needing to be in constant view of a door or other escape route. In addition to the fears themselves, the term agoraphobia is often used to refer to avoidance behaviors that sufferers often develop. For example, following a panic attack while driving, someone suffering from agoraphobia may develop anxiety over driving and will therefore avoid driving. These avoidance behaviors can often have serious consequences.

Phobias: Phobic disorders are the single largest category of anxiety disorders. It includes all cases in which fear and anxiety is triggered by a definite stimulus or situation. Between 5% and 12% of the population worldwide, suffer from phobic disorders. Sufferers typically anticipate terrifying consequences from encountering the object of their fear, which can be anything from an animal to a location to a bodily fluid to a particular situation. Sufferers understand that their fear is not proportional to the actual potential danger but still are overwhelmed by the fear.

Social Anxiety Disorder : Social anxiety disorder (SAD; also known as social phobia) defines an intense fear and avoidance of negative public scrutiny, public embarrassment, humiliation, or social interaction. This anxiety can be specific to particular social situations for e.g. as public speaking. More classically, it is experienced in most or all-social interactions. Social anxiety often manifests specific physical symptoms, including blushing, sweating, and difficulty in speaking. As with all phobic disorders, those suffering from social anxiety often will try to escape from the source of their anxiety. However, in the case of social anxiety this is particularly problematic. In severe cases can lead to complete social isolation.

Obsessive Compulsive Disorder

Obsessive–compulsive disorder (OCD) is a type of anxiety disorder primarily characterized by repetitive obsessions which are distressing, persistent, and intrusive thoughts or images and compulsions such as urges to perform specific acts or rituals. It affects roughly 3% of the population worldwide.

The OCD thought pattern may be likened to superstitions. Often the process is entirely illogical. In addition, in many cases, the compulsion is entirely inexplicable, simply an urge to complete a ritual triggered by nervousness.

Post-Traumatic Stress Disorder

Post-traumatic stress disorder (PTSD) is an anxiety disorder that results from a traumatic experience. Post-traumatic stress can result from an extreme situation, such as combat, natural disaster, rape, hostage situations, child abuse, bullying, or even a serious accident. It can also result from long-term (chronic) exposure to a severe stressor. Common symptoms include hyper vigilance, flashbacks, avoidant behaviors, anxiety, anger and depression. There are a number of treatments that form the basis of the care plan for those who are suffered with PTSD. Such treatments include cognitive behavioral therapy (CBT), psychotherapy and support from family and friends.

Acute Stress Disorder

Acute stress disorder is characterized by the development of severe anxiety, dissociative, and other symptoms that occurs within one month after exposure to an extreme traumatic stressor, e.g. witnessing a death or serious accident. As a response to the traumatic event, the individual develops dissociative symptoms. Individuals with Acute Stress Disorder have a decrease in emotional responsiveness, often finding it difficult or impossible to experience pleasure in previously enjoyable activities, and frequently feel guilty about pursuing usual life tasks.

A person with acute stress disorder may experience difficulty concentrating, feel detached from their bodies, experience the world as unreal or dreamlike, or have increasing difficulty recalling specific details of the traumatic event.

Anxiety Disorder Due to a General Medical Condition

In this, anxiety disorder due to a general medical condition, the individual experiences anxiety that causes clinically significant distress or impairment in functioning. This anxiety must represent a direct physiological, not emotional, consequence of a general medical condition.

Symptoms can include prominent, generalized anxiety symptoms, Panic Attacks, or obsessions or compulsions. There must be evidence from the history, physical examination, or laboratory findings that the disturbance is the direct physiological consequence of a general medical condition. The disturbance is not better accounted for mental disorder, such as adjustment disorder with Anxiety, in which the stressor is the general medical condition. The diagnosis is not made if the anxiety symptoms occur only during the course of a delirium. The anxiety symptoms must cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.

In determining whether the anxiety symptoms are due to a general medical condition, the clinician must first establish the presence of a general medical condition. Further, the clinician must establish that the anxiety symptoms are etiologically related to the general medical condition through a physiological mechanism. A careful and comprehensive assessment of multiple factors is necessary to make this judgment. Although there are no infallible guidelines for determining whether the relationship between the anxiety symptoms and the general medical condition is etiological, several considerations provide some guidance in this area. One consideration is the presence of a temporal association between the onset, exacerbation, or remission of the general medical condition and the anxiety symptoms. A second consideration is the presence of features that are atypical of a primary anxiety disorder (e.g., atypical age at onset or course, or absence of family history). Evidence from the literature that suggests that there can be a direct association between the general medical condition in question and the development of anxiety symptoms may provide a useful context in the assessment of a particular situation.

Anxiety Disorder not Otherwise Specified

This group comprises the disorders with prominent anxiety or phobic avoidance and that they do not meet criteria for any specific Anxiety Disorder, Adjustment Disorder with Anxiety, or Adjustment Disorder with Mixed Anxiety and Depressed Mood. The following are the examples of this group.

1. Mixed anxiety-depressive disorder: clinically significant symptoms of anxiety and depression, but the criteria are not met for either a specific Mood Disorder or a specific anxiety disorder

2. Clinically significant social phobic symptoms that are related to the social impact of having a general medical condition or mental disorder e.g., Parkinson's disease, dermatological conditions, Stuttering, Anorexia Nervosa, Body Dysmorphic Disorder)

3. Situations in which the disturbance is severe enough to warrant a diagnosis of an anxiety disorder but the individual fails to report enough symptoms for the full criteria for any specific anxiety disorder to have been met. The examples are an individual who reports all of the features of Panic Disorder without Agoraphobia except that the panic attacks are all limited-symptom attacks.

4. Situations in which the clinician has concluded that an anxiety disorder is present but is unable to determine, whether it is primary, or due to a general medical condition, or substance induced.

Separation Anxiety

Separation anxiety disorder (SepAD) is the feeling of excessive and inappropriate levels of anxiety over being separated from a person or place. Separation anxiety is a normal part of development in babies or children, and it is only when this feeling is excessive or inappropriate that it can be considered a disorder. Separation anxiety disorder affects roughly 7% of adults and 4% of children, but the childhood cases tend to be more severe; in some instances, even a brief separation can produce panic.

Situational Anxiety

Situational anxiety is caused by new situations or changing events. It can also be caused by various events that make that particular individual uncomfortable. Its occurrence is very common. Often, an individual will experience panic attacks or extreme anxiety in specific situations. A situation that causes one individual to experience anxiety may not affect another individual, at all. For example, some people become uneasy in crowds, tight spaces, so standing in a tightly packed line, say at the bank, or a store register, may cause them to experience extreme anxiety, possibly a panic attack. Others, however, may experience anxiety when major changes in life occur, such as entering college, getting married, having children, etc.

Anxiety Disorders In Children

Children may experience anxiety disorders similar to adults.

A common anxiety disorder in children is school phobia, which in some cases can be a type of separation anxiety. Sometimes the anxiety has no obvious cause. In other instances, the child may experience maltreatment from classmates, or even a teacher. They could also be stressed from the workload they are given. School phobia may also be a form of social phobia, also known as social anxiety. Children with this disorder may avoid speaking in front of their classmates or meeting new people. Typically, social phobia in children it is caused by some traumatic event, such as not knowing an answer when called on in class.

Aetiology of Anxiety Disorders

1. Drugs: alcohol abuse; Caffeine, alcohol and benzodiazepine dependence; exposure to organic solvents in the work environment;
2. Stress: Life stresses such as financial worries or chronic physical illness;
3. Genetics: GAD runs in families and is six times more common in the children of someone with the condition.
4. Persistence of anxiety:
5. Evolutionary mismatch

Epilepsy

1. A seizure is any clinical events caused by an abnormal electrical discharge in the brain, whilst epilepsy is the tendency in have recurrent seizures.
2. Epilepsy should be regarded as a symptom of brain disease rather than a disease itself.
3. A single seizure is not epilepsy but an indication for investigation, and medication should generally be with held until recurrent seizures occur.
4. The recurrence rate after a first seizure is about 70% within the first year, and most recurrent attacks occur within a month or two of the first.
5. Further seizures are loss likely if a trigger factor is definable and avoidable.
6. In some conditions epilepsy is the only feature of underlying disease, whilst in others epilepsy is just one of the manifestations.
7. The annual incidence of new cases of epilepsy after infancy is 2070/100 000.
8. The lifetime risk of having a single seizure is about 5% whilst the prevalence of epilepsy in European countries is about 0.5%.
9. The prevalence in developing countries is up to five times higher than in developed countries.

Classification

The classification of epilepsy is best achieved by considering the clinical events (the seizure), the abnormal electrophysiology, the anatomical site of seizure genesis and the pathological cause of the problem.

Classification of epilepsy

Seizure type :

- Partial (simple or complex)
- Tonic clonic
- Partial with secondary generalisation
- Tonic
- Atonic
- Absence
- Myoclonic

Physiology (EEG)

- Focal spikes/sharp waves Anatomical site
- Generalised spike and wave
- Cortex
  Temporal
  Frontal
  Parietal
  Occipital
- Generalised (diencephalon)
- Multifocal

Pathological cause :

- Genetic
- Developmental
- Tumours
- Trauma
- Vascular
- Infections
- Inflammation
- Metabolic
- Drugs, alcohol and toxins
- Degenerative

A. Primary generalised epilepsy

- The primary of idiopathic epilepsis makes up some 10% of all epilepsis, including some 40% of those with tonic clonic seizures
- Onset is almost always in childhood or adolescence.
- No structural abnormally is present and there is often a substantial genetic predisposition.
- Some like childhood absence epilepsy, are relatively uncommon, whilst others, like juvenile with epilepsy

The more common varieties of primary generalised epilepsy along with their clinical features and management.

Primary generalised epilepsies

	Incidence	Age of onset	Type of	EEG features Seizure	Provoking	Treatment factors	Prognosis
Childhood absences epilepsy	6-8/100000	4-8 yrs	Frequent brief absence	3/s spike and wave	Hyperven-tilation, fatigue	Ethosu-ximide Sodium valproate	40% develop GTCS, 80% remit in adulthood
Juvenile absence epilepsy	1-2/100000	10-15 yrs	Less frequent absences than childhood absence	Poly-spike and wave	Hyperven-tilation, sleep deprivation	Sodium valproate	80% develop GTCS, 80% seizure-free in adulthood
Juvenile myoclonic epilepsy	25-50/100 000	15-20 yrs	GTCS, absences morning myoclonus	Poly-spike and wave, photosen-sitivity	Sleep deprivation, alcohol withdrawal	Sodium valproate	90% remit with AEDs but relapse if AED withdrawn
GTCS on awakening	Common	10-25 yrs	GTCS, sometimes myoclonus	'Spike and Wave on waking and sleep onset	Sleep deprivation	Sodium valproate	65% Controlled with AEDs but relapse off treatment

(GTCS = generalised tonic clonic seizure; AED = anti-epileptic drug)

B. Partial epilepsy

Partial seizures may arise from any disease of the cerebral cortex, congenital or acquired and frequently generalise with the exceptions of a few idiopathic partial epilepsies of benign outcome in childhood, the presence of partial seizures signifies the presence of focal cerebral pathology.

Causes of partial seizures

Idiopathic

- Benign Rolandic epilepsy of childhood
- Benign occipital epilepsy of childhood

Focal structural lesions

Genetic

- Tuberous sclerosis
- Neurofibromatosis
- von Hippel-Lindau disease
- Cerebral migration abnormalities

Infantile hemiplegia

Dysembryonic

- Cortical dysgenesis
- Sturge-Weber syndrome

Mesial temporal sclerosis (associated with febrile convulsions)

Cerebrovascular disease

- Intracerebral haemorrhage
- Arteriovenous malformation
- Cerebral infarction
- Cavernous haemangioma

Tumours (primary and secondary)

Trauma (including neurosurgery)

Infective

- Cerebral abscess (pyogenic)
- Subdural empyema
- Toxoplasmosis
- Encephalitis
- Cysticercosis
- Human immunodeficiency virus (HIV)
- Tuberculoma

Inflammatory

- Sarcoidosis
- Vasculitis

Secondary generalised epilepsy

1. Generalised epilepsy may arise from spread of partial seizures due to structural disease, or may be secondary to drugs or metabolic disorders.
2. Epilepsy presenting in adult life is almost always secondary generalised, even if there is no clear history of a partial seizure before the onset of a major attack (an 'aura').

Causes of secondary generalised epilepsy

Secondary generalisation from partial seizures

- All causes of partial seizures

Genetic

- Inborn errors of metabolism
- Storage diseases
- Phakomatoses (e.g. tuberous sclerosis)

Cerebral birth injury

Hydrocephalus

Cerebral anoxia

Drugs

- Antibiotics: penicillin, isoniazid, metronidazole
- Antimalarials: chloroquine, mefloquine
- Ciclosporin
- Cardiac anti-arrhythmics: lidocaine, disopyramide
- Psychotropic agents: phenothiazines, tricyclic antidepressants, lithium
- Amphetamines (withdrawal)

Alcohol (especially withdrawal)

Toxins

- Heavy metals (lead, tin)
- Organophosphates (sarin)

Metabolic disease

- Hypocalcaemia
- Hypoglycaemia
- Hyponatraemia
- Renal failure
- Hypomagnesaemia
- Liver failure Infective
- Meningitis
- Post-infectious encephalopathy

Inflammatory

- Multiple sclerosis (uncommon)
- SLE

Diffuse degenerative diseases

- Alzheimer's disease (uncommonly)
- Creutzfeldt-Jakob disease (rarely)

Pathogenesis

1. The occurrence of epilepsy in those over the age of 60 is frequently inhibitory transmitter gamma-aminobutyric acid (GABA) is particularly important in this role, and drugs that block GABA receptors provoke seizures.
2. Conversely, excessive stimulation by excitatory neurotransmitters, such as acetylcholine, glutamate and aspartate, provoke seizure activity.
3. It is likely that both reduction of inhibition and

excessive excitation play a part in the genesis of most seizures.

4. The cerebral cortex in epilepsy exhibits hypersynchronous, repetitive discharges involving large groups of neurons, intracellular recordings of which demonstrate burst of high-frequency action potentials associated with a reduction in the transmembrane potential (paroxysmal depolarization shift).

Clinical features

1. When taking a history from a patient with suspected seizure, it is important to clarify which type of attack is occurring, bearing in mind that there may be more than one occurring in the same patient at different times.
2. Sometimes specific trigger factors can be identified namely
   - Sleep deprivation
   - Alcohol (particularly withdrawal)
   - Recreational drug misuse
   - Physical and mental exhaustion
   - Flickering lights, including TV and computer screens (primary generalised epilepsies only)
   - Intercurrent infections and metabolic disturbances
   - Uncommonly: loud noises, music, reading, hot baths

Tonic clonic seizures

1. A tonic clonic seizure may be preceded by a partial seizure (the'aura') which can take various forms
2. The patient then goes rigid and becomes unconscious, falling down heavily if standing and often sustaining injury.
3. During this phase respiration is arrested and central cyanosis may occur.
4. After a few moments, the rigidity is periodically relaxed, producing clonic jerks.
5. Some patients do not have a clonic phase and the rigidity is replaced by a flaccid state of deep coma which can persist for some minutes.
6. The patient then gradually regains consciousness, but is in a confused and disorientated state for half an hour or more after regaining consciousness.
7. Full memory function may not be recovered for some hours.
8. During the attack, urinary incontinence and tongue-biting may occur.
9. A severely bitten, bleeding tongue after an attack of loss of consciousness is pathognomonic of a generalised seizure.
10. After a generalised seizure the patient usually feels terrible, may have a headache and will often want to sleep.

Complex partial seizures

1. Complex partial seizures involve episodes of altered consciousness without the patient collapsing to the ground, especially if they arise from the temporal or, less frequently, the frontal lobe.
2. Patients stop what they are doing and stare blankly, often blinking repetitively, making smacking movements of their lips, or displaying other automatisms, such as picking at their clothes.
3. After a few minutes, consciousness returns but the patient may be muddled and feel drowsy for a period of up to an hour.
4. Immediately before such an attack the patient may report alteration of mood, memory or perception (such as undue familiarity-deja vu- or unreality-jamais vu), or complex hallucinations involving sound, smell, taste, vision, emotion (fear, sexual arousal) or visceral sensations (nausea, epigastric discomfort)

Simple partial' seizure

1. Changes in memory or perception occur without subsequent alteration in awareness, the seizure is said to be a 'simple partial' seizure.
2. Complex partial seizures arising from the anterior parts of the frontal lobe may produce bizarre behaviour patterns including limb posturing, sleep walking, or even frenetic ill-

directed motor activity with incoherent screaming.
3. These can sometimes be very difficult to distinguish from psychogenic attacks (which are more common).
4. However, abruptness of onset and relative brevity may help point to frontal seizures, as may a tendency to occur out of sleep.

Absence seizures

1. Absence seizures (petit mal) always start in childhood.
2. The attacks can be mistaken for complex partial seizures but are shorter in duration
3. Occur much more frequently (20-30 times a day) and are not associated with post-ictal confusion.
4. Absence attacks are caused by a generalised discharge that does not spread out of the hemispheres and so does not cause loss of posture.

Atonic seizures

1. These are seizures involving brief loss of muscle tone, usually resulting in heavy falls with or without loss of consciousness.
2. They only occur in the context of epilepsy syndromes which involve other forms of seizure. They are not a cause of collapse in patients without epilepsy.

Partial motor seizures

1. Epileptic activity arising in the pre-central gyrus causes partial motor seizures affecting the contralateral face, arm, trunk or leg.
2. Seizures are characterised by rhythmical jerking or sustained spasm of the affected parts.
3. They may remain localised to one part, or may spread to involve the whole side.

Jacksonian seizure

1. Some attacks begin in one part of the body (e.g. mouth, thumb, great toe) and spread (march) gradually to other parts of the body; this is known as a Jacksonian seizure.
2. Attacks vary in duration from a few seconds to several hours (epilepsia partialis continua).
3. More prolonged episodes may be followed by paresis of the involved limb lasting for several hours after the seizure ceases (Todd's palsy).

Partial sensory seizures

1. Seizures arising in the sensory cortex cause unpleasant tingling or 'electric' sensations in the contralateral face or limbs.
2. A spreading pattern similar to a Jacksonian seizure may occur, the abnormal sensation spreading over the body much faster (in seconds) than the march of a migrainous focal sensory attack, which spreads over 20-30 minutes.

Versive seizures

1. A frontal epileptic focus may involve the frontal eye field, causing forced deviation of the eyes and sometimes turning of the head to the opposite side.
2. This type of attack often generalises to become a tonic clonic seizure.

Partial visual seizures

1. Occipital epileptic foci cause simple visual hallucinations such as balls of light or patterns of colour.
2. Formed visual hallucinations of faces or scenes arise more interiorly in the temporal lobes.

Investigations

Are the attacks truly epileptic?

- Ambulatory EEG
- Videotelemetry

From where is the epilepsy arising?

- Standard EEG
- Sleep EEG
- EEG with special electrodes (foramen ovale, subdural)

What is the cause of the epilepsy?

Structural lesion?

- CT
- MRI

Metabolic disorder?

- Urea and electrolytes

- Liver function tests
- Blood glucose
- Serum calcium, magnesium Inflammatory or infective disorder?
- Full blood count, erythrocyte sedimentation rate (ESR), C-reactive protein (CRP)
- Chest X-ray
- Serology for syphilis, HIV, collagen disease
- CSF examination

Indications for brain imaging in epilepsy

- Epilepsy starting after the age of 20 years
- Seizures having focal features clinically
- EEG showing a focal seizure source
- Control of seizures difficult or deteriorating

Epilespsy in old age

- Incidence and prevalence: late-onset epilepsy is very common and the annual incidence in those over 60 years is rising.
- Fits and faints: the features that usually differentiate these may be less definitive than in younger patients.
- Complex partial status epilepticus: can present as confusion in the elderly.
- Cerebrovascular disease is the underlying cause of seizures in 30-50% of patients over the age of 50 years. A seizure may occur with an overt stroke or with occult vascular disease.
- Anti-epileptic drug regimens: keep as simple as possible and take care to avoid interactions with other drugs being prescribed.
- Carbamazepine-induced hyponatraemia: increases significantly with age; particularly important in patients on diuretics or those with heart failure.

MCQs

1. For the diagnoses of depression, symptoms must be present for at least
   a. 2 weeks b. 1 month
   c. 2 months d. 6 months
2. Anxiety Neurosis is diagnosed by the following features
   a. Excessive, uncontrollable worry
   b. Excessive, uncontrollable worry + 3 additional anxiety symptoms most days, for atleast 6 months.
   c. Excessive, uncontrollable worry + 1 additional anxiety symptoms, for at least 2 weeks.
   d. Persistent feelings of sadness, hopelessness+ 1 additional anxiety symptoms, for at least 2 weeks
3. Recurrentsuicidal ideation seen in
   a. Anxiety Disorders
   b. Bipolar Disorder
   c. Depression
   d. Post-Traumatic Stress Disorder

Answer

1. a	2. b	3. c